
02452 Machine Learning - Project 2

Supervised learning: Classification and regression

Andreas Kruse Svenningsen

s253844@student.dtu.dk

Sebastian In Soo Nygaard Wærling

s254120@student.dtu.dk

GitHub Repository:

<https://github.com/krusand/02452-ML-Project>

Section	Subsection/Subsubsection	Main Responsible
Introduction	-	Sebastian
Regression, Part A	-	Andreas
Regression, Part B	Regression models	Sebastian
Regression, Part B	Hyperparameter tuning	Sebastian
Regression, Part B	Statistical evaluation	Sebastian
Classification	Classification models	Andreas
Classification	Hyperparameter tuning	Andreas
Classification	Statistical evaluation	Sebastian
Classification	Logistic regression experiment	Andreas
Discussion	-	Andreas
Conclusion	-	Andreas

Table 1: Responsible group member for each report section.
We worked together for the majority of work, in both code and report.



MSc, Human Centered AI
TECHNICAL UNIVERSITY OF DENMARK

November 2025

1 Introduction

Regression and classification are two central tasks within machine learning. In this report, they also play a central role as we analyze whether three regression models and three classification models can predict obesity and coronary heart disease. We tune model hyperparameters using two-level cross-validation which we also use to pick an optimal hyperparameter value for each model. Finally, we compare the models in terms of their generalization error.

The report is based on project report 1, in which we explored the *South African Heart Disease* dataset [1], to assess the feasibility of performing the regression and classification tasks. We will therefore only sparsely introduce certain project elements that are already covered by report 1.

2 Regression

2.1 Data

For regression, we model obesity as the outcome, using covariates from the dataset. We use all covariates in the dataset, meaning the *systolic blood pressure (sbp)*, *cumulative tobacco intake*, *ldl cholesterol*, *type a (psycho social stress)*, *cumulative alcohol intake*, *age*, *coronary heart disease (chd)*, *family history of chd*, and *adiposity*. From our first report, we found that obesity was heavily correlated with adiposity. Both are obesity measurements, and thus this makes sense. Furthermore, obesity was correlated with age, ldl cholesterol and sbp. Since adiposity is essentially measuring the same as obesity, we will include two different sets of covariates. One with, and one without adiposity as a covariate.

We made a preprocessing pipeline, implemented in `utils.py` as a class `Preprocessor`. This handled all the preprocessing. We made two different preprocessing pipelines. One for numeric and one for discrete features. The discrete pipeline consisted of one step, a label encoding step, to take care of the `famhist` feature. Label encoding meant to map 'Present' to 1, and 'Absent' to 0. We created a class `ModifiedLabelEncoder` for this. We do not perform one out of K coding, because the two discrete features we have in the dataset are both binary. The numeric pipeline consisted of a log transformation of the two features, alcohol, and tobacco intake. We

log-transformed these features indicated an exponential relationship because they are cumulative. We implemented this as the `LogTransformer` class, functioning as part of the preprocessing pipeline. Because $\log 0$ is undefined, we added $1e-5$ to all values. This meant that after the log transformation, values which were zero before would become a large negative number. After the log transformation, we standardized each feature to mean 0 and standard deviation 1. We do this, because the regularization adds the same weight to each feature.

2.2 Part A - Linear regression

We use a form of linear regression with a penalty term added, known as Ridge regression. We use L2 penalty because our feature set is small, and thus we don't want the L1 penalty to give sparse solutions where weights of coefficients are 0. Instead L2 penalty shrinks features towards 0. The regularization parameter introduced in the course, λ , is denoted as `alpha` in the Ridge regression class we use from `sklearn`. We test a range of values from 10^{-6} to 10^2 , with more values tested for smaller regularization strengths. Less regularization means higher bias, and more regularization means higher variance. We want to find a sweet spot between these. We use K=10 fold cross validation to test the generalization error (test error). We tested both with and without adiposity. Figure 1 shows these results.

We notice a few things from these plots. First, we see that the train error is less than the test error. This suggests that the model got fitted correctly in the cross validation, and we didn't introduce data leakage. We also notice the mean error for the model with adiposity is lower compared to the model without adiposity. This suggests adiposity is an important feature, which we will explore later. The best λ for each model is 5.42 with adiposity and 39.17 without adiposity.

Since we have log transformed and standardized features, it is more difficult to interpret the coefficients from the best ridge regression. We interpret both the coefficients given from the model with and without adiposity. To get the coefficients, we fitted the model on the whole dataset, with $\lambda = 5.42$. The most important coefficient is adiposity, with a value of 3.5. This value means that whenever we move one standard deviation towards larger adiposity, the BMI (obesity) is increased by 3.5. For

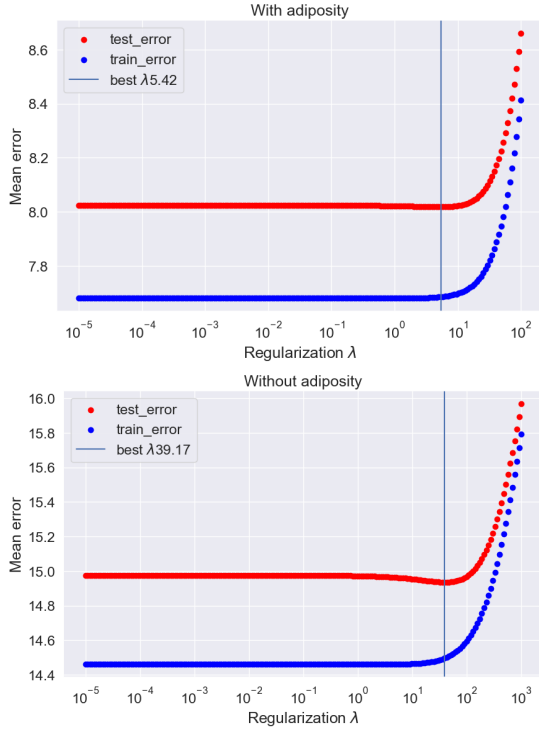


Figure 1: Regularization with and without the feature adiposity

a person 175 centimeter tall, it would correspond to gaining 11 kg in weight. Because this feature is highly predictive of obesity, most of the other features are irrelevant. Age suggests a negative relationship with obesity, but we remember that this is only valid when conditioning on adiposity. That is like saying if we know a person already has a lot of adipose tissue, then increasing their age means they become less obese. Since adiposity and obesity are more or less the same measures, we now fit a ridge regression without using adiposity as a feature. We use $\lambda = 39.17$. The result is a more balanced output, with more coefficients being important. LDL cholesterol has the coefficient 1.04, meaning an increase of 1 standard deviation in cholesterol means an increase of 1 BMI. Age (0.79) and systolic blood pressure (0.54) also increases obesity as both increases. Notably, people who have had coronary heart disease tend to be less obese. This is interesting because it could suggest that they lose weight after having had chd. Alcohol consumption is weakly, positively linked with obesity. Since we log transformed and standardized, it is harder to interpret. The alcohol coefficient was 0.2, which means a 0.2 BMI increase for every standard deviation

in log cumulative alcohol consumption. The standard deviation of alcohol after log transforming is $\sigma_{\log \text{alcohol}} = 2.2$. This means that individuals who have an alcohol intake of $e^{2.2} \approx 9$ times more than the mean alcohol intake, have a 0.2 higher BMI. This is not a lot, and therefore alcohol is probably not an important feature. Likewise for tobacco intake, people who have a tobacco intake of 6.6 times higher than the mean intake, has a 0.2 lower BMI. We also tried fitting the ridge regression to features without log transforming, but this provided the conclusion. That tobacco and alcohol were irrelevant.

Based on our initial research in report 1, we expected adiposity to be highly predictive of obesity since they are more or less measurements of the same underlying thing. We also expected ldl, age, and sbp to be predictive which was also the results. We were surprised that alcohol intake and tobacco intake was not important, but based of report 1, this was expected. Furthermore, we were surprised that people with chd were less obese.

2.3 Part B - Comparing regression models

2.3.1 Regression models

This section introduces two additional regression models to be compared to the Ridge regression model from Part A. The first model is an artificial neural network (ANN) with eight input neurons and a single hidden layer, whose dimension is controlled by the `hidden_dim` hyperparameter. The tanh function is used as the activation function to ensure non-linearity. Since the ANN is used for regression, it has only a single output neuron and uses mean squared error as its loss function. The AdamW optimizer from `torch.optim` is chosen for its computational efficiency [2] with a learning rate of 0.001. The ANN regressor is trained for 200 epochs.

The third and final regression model is a baseline regressor that always predicts the average obesity value from the training data to which it was fitted. Hence, the baseline regressor does not have any hyperparameters.

The ANN and baseline regressors are implemented as two classes, `ANNRegressor` and `BaselineRegressor`, each with a `.fit` and `.predict` method, similar to linear models from

Outer fold i	ANN		Linear regression (Ridge)		baseline
	h_i^*	E_i^{test}	λ_i^*	E_i^{test}	E_i^{test}
1	100	28.538	32.551	16.237	19.051
2	100	45.232	42.970	17.853	19.865
3	100	20.142	82.143	9.508	14.004
4	100	6.557	56.724	9.824	12.236
5	100	3.752	47.138	10.768	15.223
6	100	1.563	47.138	16.174	19.350
7	100	0.748	32.551	8.974	11.436
8	100	0.775	74.881	22.345	26.112
9	100	1.251	42.970	27.486	25.796
10	100	0.418	90.110	11.450	14.496

Table 2: E_i^{test} corresponds to the average squared error loss on the test data for each regression model and optimal hyperparameter value for each outer fold in the two-level cross-validation setup.

`sklearn.linear_model`. This made it easy to run the same code for the models and ultimately explains why we chose to implement them as classes.

2.3.2 Hyperparameter tuning

To obtain a robust and unbiased estimate of the regression models' generalization error, we are assessing their performance using two-level cross-validation with $K_1 = K_2 = 10$, where K_1, K_2 indicate the number of folds in the outer and inner loop, respectively. The generalization error is estimated by the average squared loss.

For the Ridge regression introduced in the previous subsection, we are tuning the regularization hyperparameter, λ . We are performing a thorough hyperparameter search in the interval from $\log_{10}(1e-5)$ to $\log_{10}(1000)$. The chosen values are equidistant on the log-scale, implying that we are testing more values closer to 0.

The hyperparameter to be tuned for the ANN regressor is the `hidden_dim`. We are testing the following values of `hidden_dim`: 1, 10, 20, ..., 100 with an increment of 10 from 10 to 100. The number of training epochs and the learning rate are other candidates for hyperparameters to be tuned for the ANN regressor. However, we will not go into further detail about how these hyperparameters were chosen.

The optimal hyperparameter value in each outer fold for each model type can be seen in Table 2. The optimal `hidden_dim` for the ANN regressor is con-

sistently 100. However, the average squared loss is much higher for the first three folds compared to the remaining folds, in which the ANN regressor seems to outperform the other two regression models.

Even though a total of 200 different values for λ are tested, some of the values emerged as the optimal value multiple times, e.g., 32.551. The λ values are rounded off to 3 decimal points but are still matching exactly as reflected by Table 2.

The choice of hyperparameter for each model is made by considering the results from Table 2. We choose the hyperparameter with the largest mode. If two hyperparameters have the same mode, we pick the smallest hyperparameter. Thus for ANN we pick $h = h_{1,2,3,4,5,6,7,8,9,10} = 100$ and for Ridge we pick $\lambda = \lambda_{1,7} = 32.551$.

2.3.3 Statistical evaluation

To assess whether the estimated generalization error of one regression model can be considered lower than that of another, we will use the paired t -test (Method 11.3.4 in [3]) which involves a choice of cross-validation. We will use K -fold cross-validation with $K = 10$ with the seed that was used to generate the outer folds when performing two-level cross-validation. Each model is compared pair-wise to one of the other regression models, and the corresponding p -value and 95% confidence interval are shown in Table 3.

(Model A, Model B)	p -value	95% CI
(Ridge, Baseline)	1.322e-5	$[-3.954, -1.514]$
(Ridge, ANN)	2.035e-10	$[-38.102, -20.421]$
(ANN, Baseline)	4.223e-11	$[-11.246, -6.179]$

Table 3: Results for paired t -tests of regression models.

All the reported p -values are significant, i.e., below 0.05, and the confidence intervals contain only negative values. Since the intervals represent the average difference in squared loss between model A and B, $\hat{z}$, it can be concluded that model A has a lower estimated generalization error than model B in all three tests. Meaning that the baseline regressor is outperformed by both Ridge regression and the ANN regressor. However, Ridge regression is able to outperform the ANN regressor, which is probably due to the high error during the first three outer folds for the ANN.

3 Classification

The classification problem we encounter is about predicting coronary heart disease (chd), using covariates from the dataset. We use all, except one covariate in the dataset, meaning the *systolic blood pressure (sbp)*, *cumulative tobacco intake*, *ldl cholesterol*, *type a (psycho social stress)*, *cumulative alcohol intake*, *age*, *obesity*, and *family history of chd*. We do not use adiposity to predict chd, since we found adiposity was heavily correlated with obesity.

3.1 Classification models

Our choice of classification models is similar to the regression problem. We compare logistic regression with L2 penalty, with an artificial neural network with one hidden layer of varying dimensionality. Compared to the ANN regressor, the ANN for classification uses a ReLU activation function and binary cross entropy as its loss function. Additionally, the output layer has two neurons, one for each class of chd, each with a sigmoid layer. We compare the logistic regression and ANN classifiers with a baseline model, which predicts the mode of the training set.

Like the regression models, the ANN and baseline classifiers are implemented as classes, while the logistic regression classifier is imported from `sklearn.linear_model`.

3.2 Hyperparameter tuning

For logistic regression we tune the hyperparameter C , given as $C = \frac{1}{\lambda}$, which is found in the sklearn implementation. This means that a large value of λ now corresponds to a high bias, and small values to high variance. We test the same values for λ as we did for Ridge regression. For the artificial neural network, we tune dimensionality of the hidden middle layer. We try values 1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. The baseline has no hyperparameter tuning.

To tune the parameters, we once again run two-level cross validation. The results for the classification models are shown in Table 4. From the results, we see that the ANN performs best, and has the lowest generalization error of the three models. The ANN hyperparameter hidden dimension tend to be high, which means a more complex model. The results for the logistic regression indicates the model is a bit worse compared to ANN, but a bit better than the baseline. The regularization parameter λ varies a lot, indicating uncertainty in which hyperparameter to choose. Compared to logistic regression, the baseline performs well. This could indicate the logistic regression model is not complex enough to handle the data. To curb this, we would have to engineer features further than we already have. For example we could introduce polynomial features of the original feature space, and also include interactions between features.

We are using the same method to pick the optimal hyperparameter across the outer folds as for the regression models. For ANN we would pick $h = h_{1,4,5} = 90$, for logistic regression we pick $\lambda = \lambda_2 = 0.029$. An improvement to this method would be to take into account groups of hyperparameters. For example $\lambda_1 = 0.420$, $\lambda_7 = 0.461$, $\lambda_{10} = 0.608$ are all close to each other, and therefore one could make a method which picks the mean of these. However, we did not have sufficient time to do this.

3.3 Statistical evaluation

Once again, we perform a statistical test to assess whether the performance of one model can be considered better than that of another. For the classifiers, we are particularly assessing whether the error rate of one classifier is lower than the error rate of another. The statistical test of choice is McNe-

Outer fold i	ANN		Linear regression (Ridge)		baseline
	h_i^*	E_i^{test}	λ_i^*	E_i^{test}	E_i^{test}
1	90	0.319	0.420	0.213	0.340
2	100	0.106	0.029	0.277	0.404
3	80	0.130	3.218	0.174	0.217
4	90	0.043	6.748	0.239	0.391
5	90	0.065	8.120	0.348	0.326
6	50	0.109	1.060	0.370	0.391
7	40	0.065	0.461	0.261	0.391
8	50	0.087	32.551	0.326	0.370
9	80	0.000	4.660	0.283	0.370
10	60	0.087	0.608	0.348	0.261

Table 4: E_i^{test} corresponds to the error rate on the test data for each classifier and optimal hyperparameter value for each outer fold in the two-level cross-validation setup.

mar’s test (Method 11.3.2 in [3]). The test is carried out using K -fold cross validation with the same seed used to generate the outer folds in the two-level cross-validation setup. The results of the tests can be seen in Table 5. A significance level of 0.05 is chosen, the same as for the paired t -tests.

(Model A, Model B)	p -value	95% CI
(Logistic, Baseline)	0.021	[0.011, 0.110]
(ANN, Baseline)	8.703e-46	[0.294, 0.381]
(Logistic, ANN)	1.462e-8	[-0.165, -0.082]

Table 5: Results for McNemar’s test comparing classification models.

Once again, all the tests yield significant p -values. For McNemar’s test, the confidence intervals represent θ , which is estimated by

$$\hat{\theta} = \frac{n_{12} - n_{21}}{n}$$

Thus, it is fair to assume that $n_{12} > n_{21}$ when comparing both the logistic regression and ANN classifier to the baseline, which implies that the baseline more often yields an incorrect prediction when the other classifiers are right in their predictions. This means that the baseline classifier is outperformed by both the logistic regression and ANN classifier.

When comparing the logistic regression classifier to the ANN classifier the confidence interval only contains negative values, why it can be assumed that $n_{12} < n_{21}$. The interpretation of this result is that the ANN classifier has a lower error rate than

the logistic regression classifier, and is thus our superior classification model.

3.4 Logistic regression experiment

We use the same λ parameter as explained in the hyperparameter results. Since the task is now classification, and we are predicting something different, we don’t necessarily expect the same features to be important. Indeed, there are some differences. Age is still a highly predictive feature, and the coefficient estimate shows that for every standard deviation increase in age, the relative increase in having chd is $e^{0.75} \approx 2.11$, 111%. If anybody in your family has had chd, the increase in risk for having chd is $e^{0.89} \approx 2.43$, 143%. Tobacco shows a 50% risk increase when consuming one standard deviation more of the log-transformed than previously consumed.. For example a perrson moving from the mean to +1SD. LDL cholesterol and typea are also deemed important. Notably being obese reduces your risk of chd. Thus when comparing to the task of predicting obesity, some features are seen in both, like age and ldl cholesterol. However, there are also features only predictive of obesity, and only predictive of chd. Tobacco and famhist are for example only predictive of chd.

We now go through an example of how logistic regression predicts the chd class and probability for an observation. The logistic regression has the following weights, and uses the following observation

values, found after preprocessing the data.

weights	observation
$w_{\text{intercept}} = -1.10$	
$w_{\text{obesity}} = -0.16$	$x_{i,\text{obesity}} = 0.67$
$w_{\text{alcohol}} = -0.052$	$x_{i,\text{alcohol}} = 0.27$
$w_{\text{typea}} = 0.37$	$x_{i,\text{typea}} = 0.19$
$w_{\text{tobacco}} = 0.41$	$x_{i,\text{tobacco}} = -0.49$
$w_{\text{sbp}} = 0.16$	$x_{i,\text{sbp}} = 0.28$
$w_{\text{age}} = 0.75$	$x_{i,\text{age}} = 1.38$
$w_{\text{ldl}} = 0.34$	$x_{i,\text{ldl}} = -0.16$
$w_{\text{famhist}} = 0.89$	$x_{i,\text{famhist}} = 0$

We use the above weights and observation, and input those as follows:

$$\begin{aligned}
y_i &= \sigma(w_{\text{intercept}} + w_{\text{obesity}} \cdot x_{i,\text{obesity}} \\
&\quad + w_{\text{alcohol}} \cdot x_{i,\text{alcohol}} + w_{\text{typea}} \cdot x_{i,\text{typea}} \\
&\quad + w_{\text{tobacco}} \cdot x_{i,\text{tobacco}} + w_{\text{sbp}} \cdot x_{i,\text{sbp}} \\
&\quad + w_{\text{age}} \cdot x_{i,\text{age}} + w_{\text{ldl}} \cdot x_{i,\text{ldl}} \\
&\quad + w_{\text{famhist}} \cdot x_{i,\text{famhist}}) \\
&= \sigma(-1.29 + (-0.16) \cdot 0.67 \\
&\quad + (-0.052) \cdot 0.27 + 0.37 \cdot 0.19 \\
&\quad + 0.41 \cdot (-0.49) + 0.16 \cdot 0.28 \\
&\quad + 0.75 \cdot 1.38 + 0.34 \cdot (-0.16) \\
&\quad + 0.89 \cdot 0) \\
&= \frac{1}{1 + \exp^{-0.5151}} \\
&= 0.374
\end{aligned}$$

Logistic regression predicts a 37.4% probability of being $\text{chd}=1$, meaning it would decide this was not chd ($\text{chd}=0$). The correct answer was 1, so our model is incorrect when predicting chd here. Thus it would give an error. In this case that error is a false negative, which is not a good thing when predicting medical diagnoses.

4 Discussion

We start of the discussion by discussing which features were important in each tasks. For both tasks, age was a highly predictive feature, most important in classifying chd . LDL cholesterol was also important for both, but most important in predicting obesity. We found that obesity and chd did not have

a clear relation, and both being bad predictive features of each other.

From report 1, we discussed what previous studies have found. We refer to this for more background. We focus only on comparing our classification results since that is the primary task of the dataset. Compared to previous studies, we found the same features to be important. Previous studies found Type A, family history, tobacco intake, age, cholersterol, and systolic blood pressure to be important. Systolic blood pressure was only found to be important after including whether the patient took medicine for high blood pressure. We also found these variables to be important, except systolic blood pressure. We could however not include interaction between taking medicine for systolic blood pressure since we did not have this data.

In terms of our models, both ANN and logistic/linear regressions outperformed the baseline models, for both tasks. This means our data show some good predictors, which the models picked up on. The best model in the regression task was ridge regression. This indicates the artificial neural network either did not train enough, or was not complex enough, or even overfitted. A better setup would probably be to have more hidden layers, each of smaller dimensions. Furthermore, we could have introduced early stopping for ANN. That is, we test for every epoch also the test set. If after k epochs the test error did not fall, we use the model with the best test error, and stop training the model. This would prevent overfitting.

For the classification task, the ANN classifier significantly outperformed logistic regression. Perhaps this indicates predicting chd is more complex, and thus ANN needed the extra epochs and didn't overfit.

One thing we encountered was the fact that we needed to use many epochs using standard gradient descent methods like stochastic gradient descent. This meant that we had to run about 60.000 epochs to achieve good results on the test set. This is an important factor when running $10 \times 10 \times 11 = 1100$ runs due to inner and outer folds as well as possible hyperparameter values. One fit using 60.000 epochs took about 1 minute. This meant that our hyperparameter tuning would take 1100 minutes, or about

37 hours when testing both classification and regression for ANN. We decided we did not want to wait this long, and instead turned to a different optimizer AdamW, which introduces weight decay, on top of the already improved Adam optimizer, which introduces momentum into gradient descent. Momentum introduction is simply explained as variable learning rate.

Furthermore, our dataset was small with only $n = 462$ samples. This could introduce more noise into the model fits, and more data would help curb noise by providing more information.

5 Conclusion

Regression and classification could also be used to analyze the problem of predicting obesity and coronary heart disease in the *South African Heart Disease* dataset. We found through two-level cross validation, some optimal parameters. Our ANN models preferred a larger hidden dimension. Regularization of logistic regression was more varied. For ridge regression, higher values of λ were preferred. The best model for the regression task was ridge regression with $\lambda = 32.6$. For classification, the best model was an artificial neural network with one hidden layer of $h = 90$ neurons. By analyzing the coefficients of logistic and ridge regression, we found some features which were predictive of obesity and chd. That included age and LDL cholesterol. Our analysis of important features in the classification task conformed to the important features found in previous studies of the same dataset.

6 Use of Generative AI and scripts from course

We have used generative AI to format tables and figures in LaTeX. For instance combining two figures into subfigures: Figure 1. Furthermore, we took inspiration from scripts given during the exercises.

References

- [1] R.W. Oldford. South African Heart Disease Data. Retrieved September 23, 2025 from https://great-northern-diver.github.io/loon.data/reference/SAheart.html?utm_source=chatgpt.com#author-1
- [2] I. Loshchilov and F. Hutter. 2019. Decoupled Weight Decay Regularization. <https://doi.org/10.48550/arXiv.1711.05101>
- [3] T. Herlau, M. N. Schmidt and M. Mørup. Chapter 11: Performance evaluation, in Introduction to Machine Learning and Data Mining. Technical University of Denmark. 2023. p195-220.